

SEDAR 17: South Atlantic Spanish mackerel and Vermilion Snapper

South Atlantic Spanish Mackerel

Data Workshop

Recommendations of the Life History Work Group

1. Ages provided for future assessments should be advanced when appropriate (i.e., during months when annuli are being formed) so fish can be assigned to the correct year class. If advanced ages cannot be provided, data should include assessment of otolith edge type. Classification schemes for edge type and quality of the otolith/section have been developed by the MARMAP program at SCDNR and are currently used by MARMAP and NMFS Beaufort.
2. Conduct inter-lab comparisons of age readings from test sets of otoliths in preparation for any future stock assessments.
3. Obtain adequate data to determine gutted to whole weight relationships.
4. Investigate the discard mortality of Spanish mackerel in the commercial and recreational trolling fishery, commercial gillnet fishery, and the shrimp trawl fishery.
5. To ensure more accurate estimates of t_0 , increase efforts to collect age 0 specimens for use in estimating von Bertalanffy (VB) growth parameters.

Recommendations of the Commercial Work Group

1. Need observer coverage for the fisheries for Spanish mackerel (gillnets, castnets
2. (FL), handlines, poundnets and shrimp trawls for bycatch):
 - 5-10% allocated by strata within states
 - possible to use exemption to bring in everything with no sale
 - get maximum information from fish
3. Expand TIP sampling to better cover all statistical strata
 - Predominantly from Florida and by gillnet & castnet gears
 - In that sense, we have decent coverage for lengths
4. Trade off with lengths versus ages, need for more ages (i.e., hard parts)
5. Need to address issue of fish retained for bait (undersized) or used for food by crew. (how to capture in landings)

Recommendations of the Recreational Work Group

There was insufficient time for this topic to be addressed by the workgroup during the data workshop.

Recommendations of the Indices Work Group

1. Expand existing fishery independent sampling and/or develop new fishery independent sampling of the Spanish mackerel population off the southeastern U.S.

Two ideas discussed were the following:

- Collect age samples from SEAMAP
- Fishery independent sampling of adults

2. Investigate whether catchability varies as a function of fish density and/or environmental conditions.

3. Investigate how temporal changes in migratory patterns may influence indices of abundance (for fishery dependent and fishery independent indices).

4. Investigate the possibility of using models that allow catchability to follow a random walk.

*Assessment Workshop***Recommendations of the Assessment Panel**

Comprehensive Data and Assessment Archive: A goal of the SEDAR process, as stated in several workshop Terms of Reference, is to properly document all aspects of the data employed in the assessments, the assessments themselves, and the peer review of assessment details and results. While the various workshop reports and data workbooks compile much of the information, concern has been expressed that a full compilation of data manipulations, and programs used to generate the final data used in the assessment is not available following a SEDAR cycle. The concept of a SEDAR Comprehensive Data and Assessment Workshops Archive was proposed by the SEDAR 17 Data Compiler during preparations for the DW. Though the idea was not advanced from the DW as a formal recommendation it was generally taken favorably. An archive could serve as: a single reference for anyone wishing to dig deeper into how data were processed, a reference for future assessments, a backup of final data processing programs or spreadsheets for those who develop them, and continuity in cases of personnel changes for future assessments and updates. When discussed at the AW it was recognized implementation of an archive could have benefits and costs, but that it would require more attention than SEDAR 17 AW participants could give it, and all SEDAR cooperators were not present. The AW recommends that a SEDAR-wide workgroup be convened to identify the pros and cons of a Comprehensive Data and Assessment Archive for each future SEDAR.

Independent Expert on Assessment Panel: The assessment panel recommends that for future SEDAR assessment workshops, a scientist experienced in assessment methods and

modeling (such as a CIE reviewer, or a NMFS or state person from outside the region) be provided as a workshop panelist. An independent expert can participate in discussing technical details of the methods used for SEDAR assessments, and assist in decisions related to model configuration during the workshop. In particular, the analysts believe that an independent analyst could contribute fresh information to improve the assessments.

Review and Qualification of Historic Recreational Angler Survey Reports: Pre-MRFSS catch and related effort data from south Atlantic recreational fisheries are very scarce, but are considered valuable to stock assessments, where available. Two reports of the U. S. Fish and Wildlife Service (SEDAR 17-RD13 and SEDAR 17-RD14) and one of the NMFS (SEDAR 17-RD15) characterize south Atlantic salt-water angling effort and success based on recall surveys conducted in 1960, 1965, and 1970, respectively. These references have been viewed in various ways in previous stock assessments performed through the SEDAR process. In SEDAR 2 for South Atlantic black sea bass, these data were not used explicitly in the age-structured modeling, however, with assumptions, were used to extend the time frame for application of the production modeling approach. In SEDAR 15 for South Atlantic red snapper these data were employed by the assessment panel at face value for the three survey years and to interpolate recreational landings before, between, and after survey years. In SEDAR 15 for South Atlantic greater amberjack the review panel agreed with the assessment panel that the survey estimates of recreational landings of “jacks” not be included in the assessment due in part to species identification concerns. For the present assessment the assessment panel has employed the survey data for both stocks under assessment, but considers recall bias on the part of persons surveyed to be a significant factor. Thus they chose to reduce the weight of the estimates in its base runs and explore the effect on the model through sensitivity runs.

A guiding principal of the SEDAR process is consistency in the identification and utilization of data that characterize fishery stocks under assessment and the fisheries that affect the stocks. Because the three pre-MRFSS saltwater angling survey reports have proven of value, and likely will be referenced in future stock assessments, the AW recommends they be reviewed by a group of fishery professionals. The group should include persons knowledgeable in survey design, data collection, and application of survey data to fishery stock assessments. The group’s function would be to qualify the three surveys, and others which the group may identify, and provide guidelines that further consistency in their utilization in future stock assessment conducted under the SEDAR process. The review of these reports could be coupled with a review and qualification of commercial and other data to standardize their use in stock assessments, as recommended in the SEDAR 17 data workshop reports.

Avoid Brief Workshop Interims: The panel made a recommendation against scheduling abbreviated SEDAR stock assessments. AW participants felt that an abbreviated schedule could compromise the quality of the assessment.

Review Workshop

Research Recommendations of the Review Panel

In its review of DW research recommendations the RW noted the recommendation to increase samples should be accompanied by information on the methodology to determine adequate sample sizes for both length frequency and age samples. Some recommendations for future research related to indicators of population abundance were outlined; however, for those to be useful, a clear statement of the problem, research objectives, methodology and identification of groups and/or projects that could undertake such research should be specified. The RW noted that the DW provided useful recommendations regarding life history, commercial, and indices. However, some of these recommendations need to be more specific and deadlines and personnel assignments identified. The need of a fishery independent index of the adult population was mentioned but ways forward were not spelled clearly enough.

In its review of pre-AW changes in data, the RW noted estimation of shrimp bycatch data resulted in a highly variable time-series, which was not fully justified. Lack of consistency with historical data requires clarification. Better documentation of the shrimp bycatch estimation procedure would be useful. Pre-MRFSS catch estimates are not available, and data for the period 1950 – 1980 was extrapolated from 3 data points, which raised some concern. Research into estimating historical recreational catch should continue.

As to estimation of uncertainty in the SCA model, the RW states research into better methods to include the uncertainty in landings history is recommended.

South Atlantic Vermilion Snapper*Data Workshop***Recommendations of the Life History Work Group**

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The numerous research recommendations from the DW and AW were not explicitly discussed at the RW. Individual panelists reviewed the recommendations and were in broad agreement with the suggestions. However, there is a clear need for the recommendations to be prioritized. Also, the Panel recommended that a proper statistical framework be used for the catch-at-age models. This would allow alternative parameterizations to be evaluated in terms of AIC or some other statistical criteria, and the calculation of standardized residuals (which allows the appropriateness of relative data weightings to be judged).

The AW base model estimates that over-fishing is occurring and that stock size is close to the over-fished threshold. This suggests that the next assessment should be sooner than the normal timeframe for assessment updates.